

C. Kanade's Perfect Multiples

time limit per test: 2 seconds
 memory limit per test: 256 megabytes

*We've carved those memories
 into ourselves... No matter how
 hard they were, they're the lives
 we carried out!*

— Angel Beats!

In the afterlife school, Kanade studies a peculiar number game.

She gives you two integers n and k , as well as an array a consisting of n integers, where $1 \leq a_i \leq k$ holds.

For an integer set $B = \{b_1, b_2, \dots, b_m\}$ where $1 \leq b_i \leq k$, we call it *complete* if and only if both of the following hold:

- For each $1 \leq i \leq n$, at least one divisor of a_i is contained in B ;
- For each $1 \leq j \leq m$, all positive multiples of b_j which are less than or equal to k appear in the array a at least once.

You have to find a *complete* set B with **minimum** possible size, or determine that no such set exists.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and k ($1 \leq n \leq 2 \cdot 10^5$, $1 \leq k \leq 10^9$) — the length of a and the upper bound of elements of a .

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq k$) — the elements of a .

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case:

- If no *complete* set B exists, print a single integer -1 in the only line of output.
- Otherwise:
 - First print a single integer m ($1 \leq m \leq n$) in the first line of output — the size of B . Note that you have to minimize the size of B .
 - Then output m integers $b_1, b_2, \dots, b_m$ ($1 \leq b_i \leq k$) in the second line — the set you constructed.

If there are multiple answers, you may print any of them.

Example

input

Copy

```
4
4 6
```

Codeforces Round 1068 (Div. 2)

Contest is running

01:28:13

Contestant



→ Submit?

Language: GNU G++17 7.3.0

Choose file: Choose File No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	438
Problem B	876
Problem C	1095
Problem D	1533
Problem E	1971
Problem F	2628
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:31 from the first attempt

```
3 2 4 6
5 5
1 2 3 4 5
3 6
2 3 6
1 2
2
```

output

Copy

```
2
2 3
1
1
-1
1
2
```

Note

In the first test case, $B = \{2, 3\}$ works. For $b = 2$, all multiples $\{2, 4, 6\}$ (up to $k = 6$) appear in a ; for $b = 3$, multiples $\{3, 6\}$ appear. Every a_i is divisible by 2 or 3. No single b can satisfy both conditions, so $m = 2$ is minimal.

In the second test case, $B = \{1\}$ satisfies both rules since every number is divisible by 1 and all multiples of 1 up to k appear.

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